Turnado - Adaptive Turn-Taking for Speech Models

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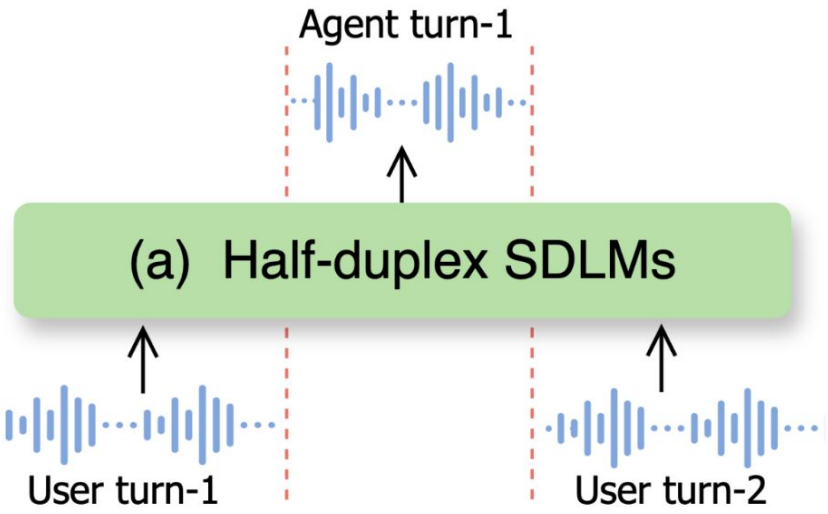
Mini Project

RQ: How well can we adapt turn-taking policies by applying a lightweight adapter to a full-duplex speech-to-speech system?

…using interpolated adapters to vary the policy between distinct turn-taking styles.

How is Spoken Conversation modelled?



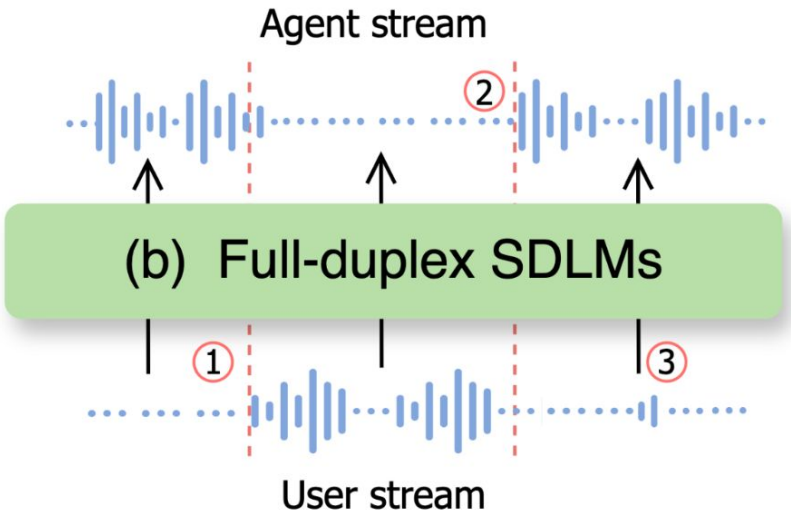
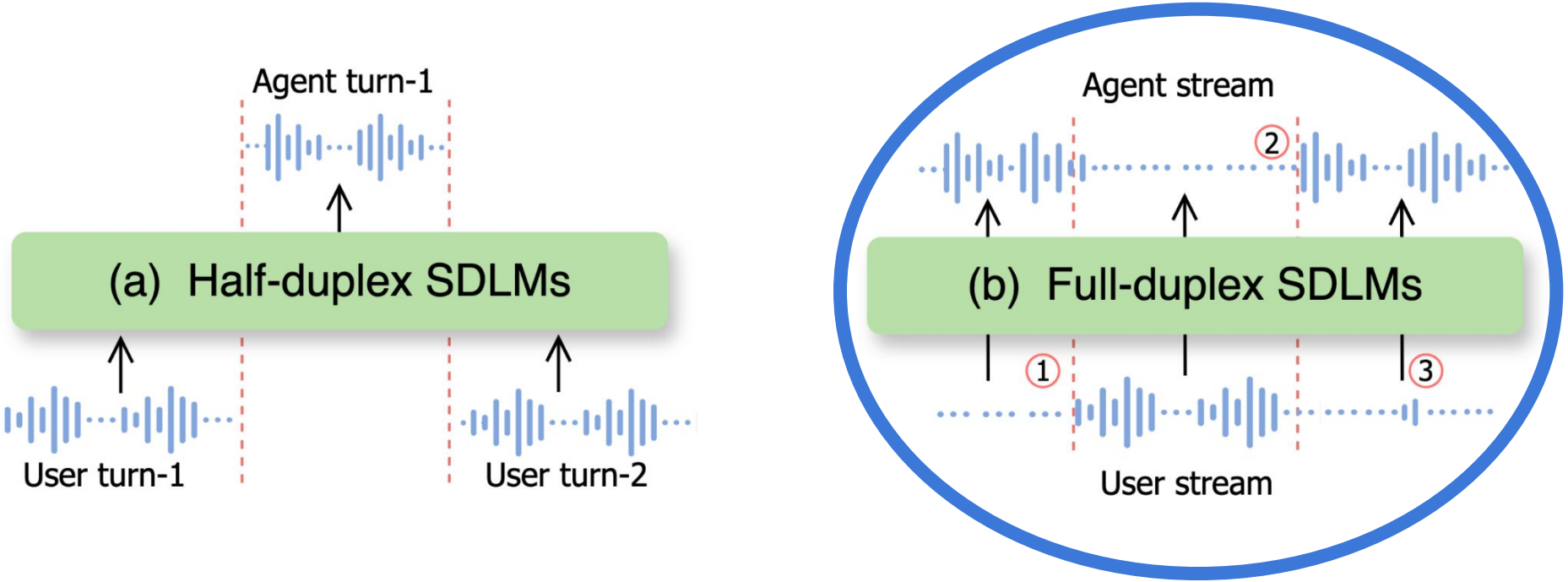


image taken from: https://openreview.net/pdf?id=QbLbXz8Idp

How is Spoken Conversation modelled?



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Data

Data

1. Select Conversation Dataset

Fisher dataset, audio recordings of conversations with time stamped transcripts

```
24.46 26.32 B: oh we used to have a dog
25.16 27.60 A: [laughter] oh really what type
26.99 28.68 B: yeah it was a beagle
```

2. Score Interruptiveness

Using timestamps, compute 0-1 score for how “interruptive” is each speaker’s turn-taking behavior.

3. Partition Dataset

Split into "low" vs "high" interruptiveness subsets by score percentile (e.g. bottom 25% vs top 25%).

*Score = (speaker starts while O is speaking) / (total O turns)

What model?

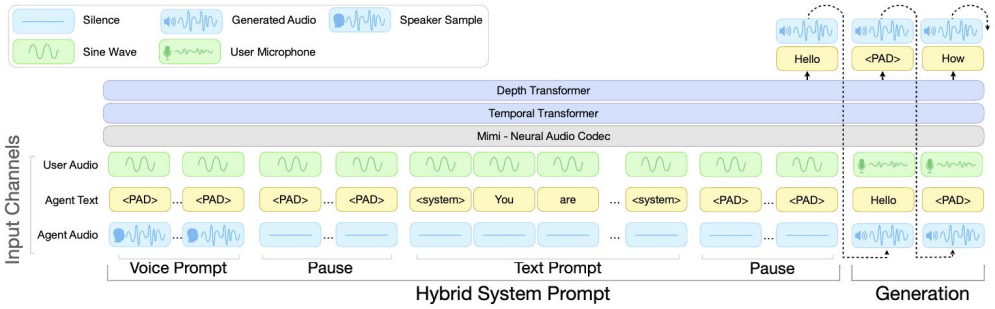
What model?

- Moshi (Kyutai)
- Hibiki (Kyutai)
- **PersonaPlex (NVIDIA / fal.ai)**
- personaplex/realtime
- LSLM (ByteDance)
- Hertz-dev
- SALMON-omni
- ....

Why PersonaPlex?

SOTA turn-taking ability

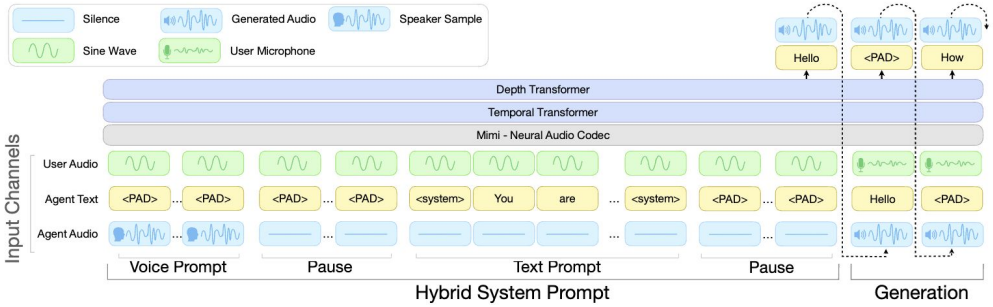
open-weights

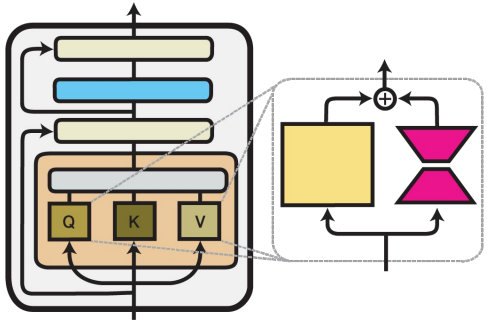


trained for Full-duplex conversation

Training

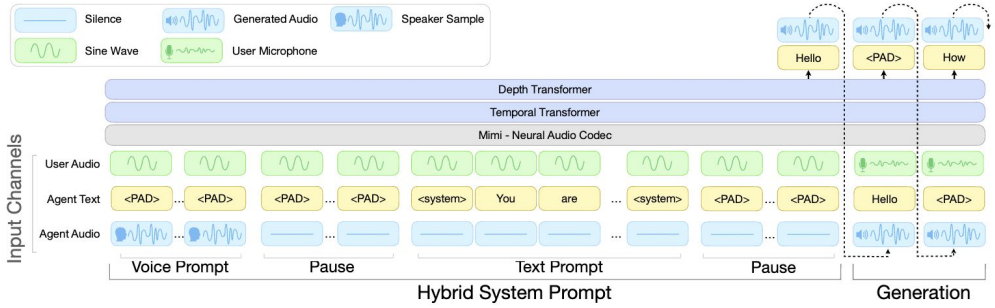
Backbone model

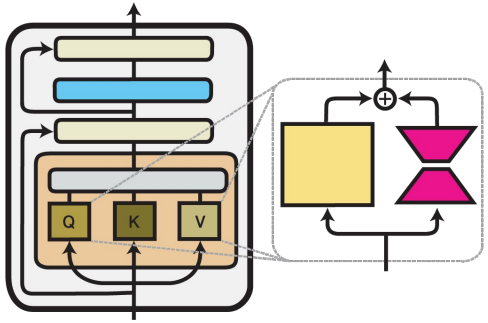




images taken from:
https://adapterhub.ml/blog/2022/09/updates-in-adapter-transformers-v3-1/

Backbone model





Adapter decision placement

ver I

Temporal transformer

ver II

Depth transformer

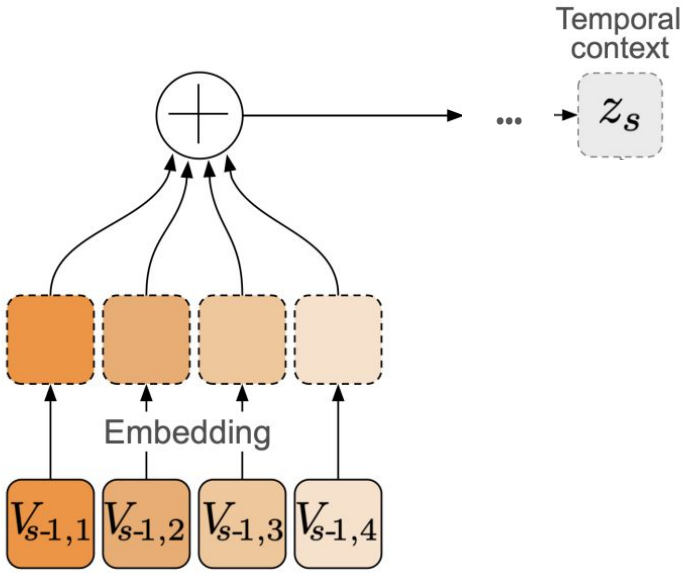
+

Temporal transformer

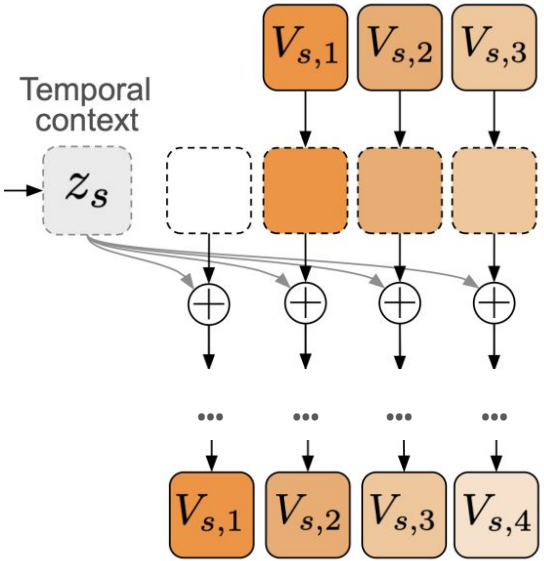
images taken from: Moshi paper
https://arxiv.org/pdf/2410.00037

Adapter decision placement

Temporal transformer

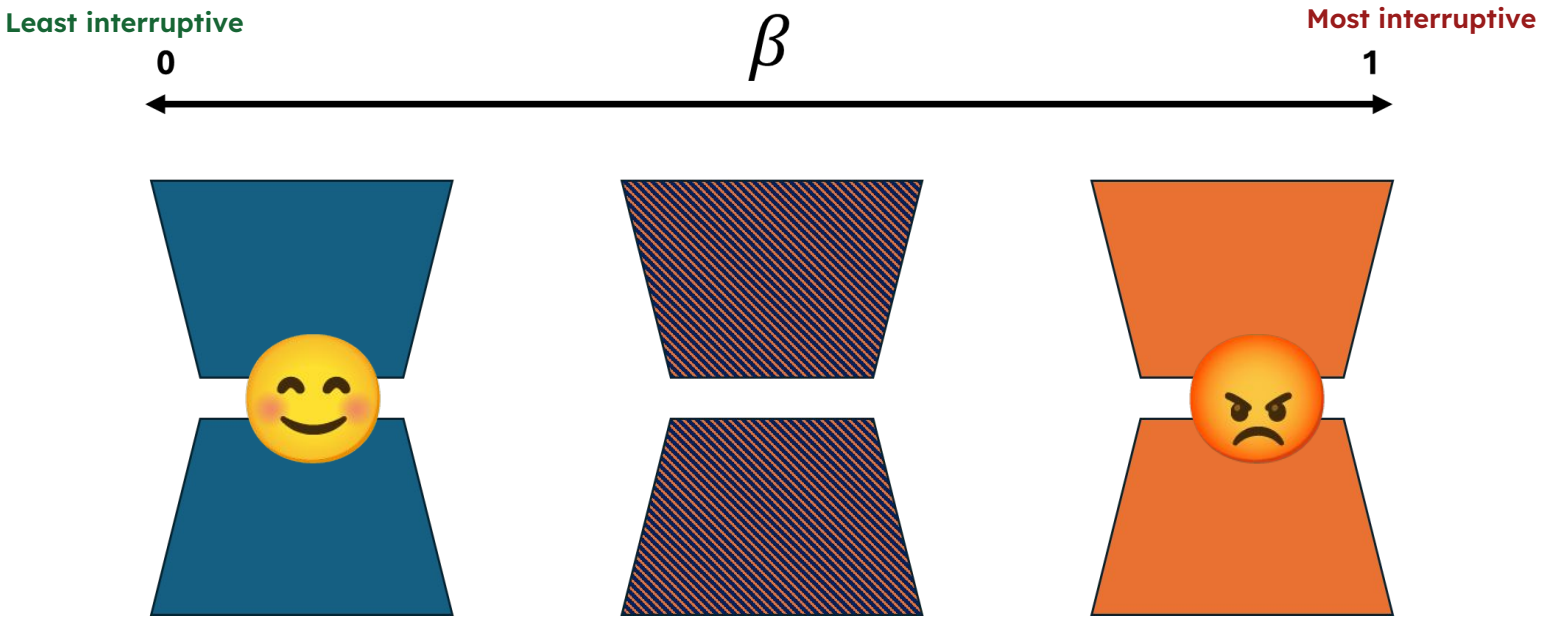


Depth transformer



Adapter Interpolation

Beta flexible at inference time



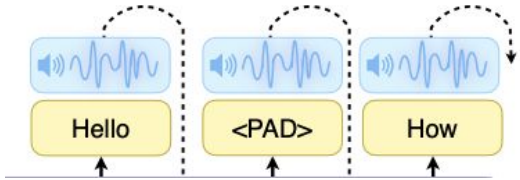
Adapter Interpolation

Is linear interpolation expressive enough?

Evaluation

→ How well does the model reflect the ***interruptiveness*** measured for the **chosen policy**.

Model output: includes text tokens with silence padding <PAD>,
but could be adjusted to <YIELD>, <PAUSE>, <SPEAK>, <INTERRUPT>.



<YIELD> <PAUSE> <SPEAK> <INTERRUPT>

Measure **closeness** between interruptive scores from model output and data set.

Further analysis: *model latency*, Full-Duplex-Bench v1.5 (Lin et al., 2026): *user interruption, user backchannel*

Thank Y–

No, thank YOU!

Appendix

Evaluation

Evaluation

Decide how to evaluate (metric, how close to the data) see if we capture the characteristics.

- Evaluating the turn-taking results of the adapted model
- 1) What do the results look like?
    - Text tokens with speech tokens simultaneously
    - Text tokens include <PAD> but could be adjusted to use <YIELD>, <PAUSE>, <SPEAK>, <INTERRUPT>
- Look at Full-Duplex-Bench → we could apply how they evaluate the data to our dataset
    - https://arxiv.org/pdf/2507.23159 FDB v1.5
    -
- Need a *closeness measurement* between turn-taking policy of the fine-tuned model and the data
    - Closeness between: measurement over model output compared to measurement over dataset output
        - (how close does the measured policy of the m

- Metric: interruptiveness
- If more time: would also consider 1) interruptiveness 2) latency
    - Interruptiveness: number of interruptions
    - Latency: length of pause

→ control for certain aspect of turn-taking (interruptiveness)
→ then evaluate on other aspects of turn-taking:

1. Model Interruption
2. Model Latency
3.
4. (below Full-Duplex-Bench)
5. User Interruption
6. User Backchannel

Adapter Interpolation

Proposed kernel-based interpolation:

$$4\beta(1-\beta)\cdot \underbrace{v \otimes w}_{\text{kernel}} \cdot (\beta \cdot ADAP_1 + (1-\beta)\cdot ADAP_2) + \frac{\beta(1-\beta)}{4} \cdot \underbrace{(\beta \cdot ADAP_1 + (1-\beta)\cdot ADAP_2)}_{\text{linear interpolation}}$$

Ensures rotational/translational interpolation between the adapters

v, w learnable vectors, trained analogously to KronA

Adapter Interpolation

Linear Interpolation between similar adapters:

Wu, Chengyue, et al. "Pi-Tuning: Transferring Multimodal Foundation Models with Optimal Multi-task Interpolation." *International Conference on Machine Learning*. PMLR, 2023.

Linear interpolation for user preferences of LLMs:

Kangaslahti, Sara, and David Alvarez-Melis. "Continuous Language Model Interpolation yields Dynamic and Controllable Text Generation." *Transactions on Machine Learning Research* 2025.

Weight averaging of adapters:

Chronopoulou, Alexandra, et al. "Adaptersoup: Weight averaging to improve generalization of pretrained language models." *Findings of the Association for Computational Linguistics: EACL 2023*. 2023.